



A Project Report

on

CAMPUS RENT: A SECURE AND SCALABLE WEB PLATFORM FOR RESOURCE RENTAL IN ACADEMIC ENVIRONMENTS

submitted as partial fulfillment for the award of

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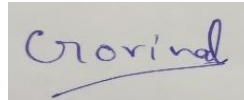
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DECLARATION

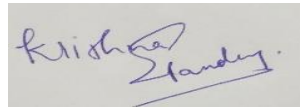
We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled “Campus Rent: A Secure and Scalable Web Platform For Resource Rental In Academic Environments” which is submitted by Prakhar Patel, Govind Dhakarey and Krishna Nand Pandey in partial fulfilment of the requirement for the award of degree B. Tech. in Department of CSE(AI) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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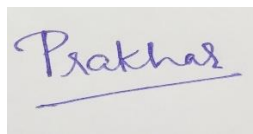
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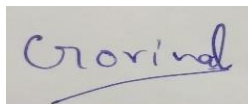
We also take the opportunity to acknowledge the contribution of **Dr. Rekha Kashyap, Dean of the Department of Computer Science & Engineering (AI/AI & ML), KIET, Ghaziabad**, for her full support and assistance during the development of the project.

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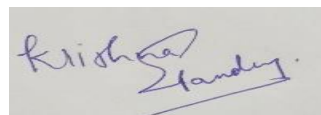
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ABSTRACT

In today's time many students prefer to use items only when needed instead of buying them. In a college environment, students often require things like books, laptops, tools and furniture for a short period. Buying these items every time can be expensive and borrowing through friends or social media is not always reliable or safe.

To solve this problem this project introduces Campus Rent a web-based platform that allows students to rent and share items within their college. The system provides a single place where students can list items they want to rent out and also find items they need. It creates a safe and organized environment by ensuring that all users are verified.

The platform is developed using the MERN stack which includes React for the frontend, Node.js and Express for the backend and MongoDB for data storage. It includes important features such as user registration, login, item listing, searching, booking and secure online payment. These features make the system easy to use and efficient.

The Campus Rent system helps students save money, reduces waste and makes better use of available resources. It also promotes a culture of sharing and cooperation among students.

Overall this project provides a simple, secure and practical solution for renting items within a campus and improves the overall student experience.

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LIST OF ABBREVIATIONS

JWT	JSON Web Token
JSON	JavaScript Object Notation
CRUD	Create, Read, Update, Delete
DB	Database
UI	User Interface
API	Application Programming Interface
AI	Artificial Intelligence

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO PROJECT

In recent years there has been a noticeable shift in the way people perceive ownership. Instead of purchasing items for permanent use individuals are increasingly adopting the concept of using resources only when required. This change is primarily driven by the rising cost of living, limited usage duration of many products and growing awareness about reducing waste. Renting therefore has emerged as a practical, economical and sustainable alternative particularly for students who often need items only for a short period of time.

Within a college environment, students regularly require a diverse range of items such as academic books, laptops, calculators, sports equipment, furniture and various tools needed for projects and assignments. However buying all these items is not always easy due to financial constraints and their temporary usage. As a result students often depend on borrowing from friends, seniors, or acquaintances. In some cases, they use informal channels such as WhatsApp groups, social media platforms or personal networks to find the required items.

Although these methods may appear convenient, they come with several limitations and risks. There is usually no proper system to verify users, which creates issues related to trust and reliability. Information about items is often incomplete or unclear leading to confusion and misunderstandings. Additionally there is no structured mechanism to handle transactions, track rentals or ensure the safe return of items. These shortcomings can result in problems such as miscommunication, delays, unfair pricing, or even fraud, making the overall process inefficient and unreliable.

To address these challenges the Campus Rent platform is proposed as a structured and secure solution. The system is designed to provide a dedicated digital space where students within a campus can list, discover and rent items in an organized manner. It ensures that all users are authenticated, thereby creating a trusted community. Each item listed on the platform includes detailed information such as description, pricing and availability, which helps users make informed decisions.

Furthermore the platform incorporates features to manage rental transactions effectively, ensuring transparency and accountability. By streamlining the entire process—from listing items to completing rentals—the system eliminates the inefficiencies of informal methods and provides a reliable alternative for students.

The primary goal of this project is to build a digital platform that bridges the gap between students who have items available for rent and those who need them. In addition to solving practical problems the system also promotes efficient resource utilization, reduces unnecessary expenditure and encourages a culture of sharing within the campus community. By supporting

a more sustainable and collaborative approach Campus Rent aims to enhance the overall student experience and contribute to a smarter way of managing resources.

1.2 PROJECT DESCRIPTION

Campus Rent is a full-stack web application developed to make the process of renting items within a college campus simple, secure and efficient. The platform is built using the MERN stack, which includes React.js for the frontend, Node.js and Express.js for the backend and MongoDB for managing the database. These technologies work together to provide a smooth and responsive user experience.

The system allows students to create an account, log in securely and start exploring available items for rent. Users can list their own items by adding details such as name, category, price, description and availability. Other students can browse these listings, select the items they need, and book them for a specific duration.

To ensure security the platform uses authentication techniques like JWT (JSON Web Token) and encrypts user passwords. This helps in protecting user data and restricting unauthorized access. The system also includes proper validation checks to make sure that all the information entered by users is accurate and reliable.

One of the key features of the platform is the integration of an online payment system using Razorpay. It allows users to make payments through different methods such as UPI, cards or net banking. The payment process is verified on the server side to ensure that transactions are safe and genuine.

Additionally an admin panel is included to monitor activities on the platform. The admin can manage users, review item listings and handle any issues or misuse, which helps maintain a safe environment for everyone.

Overall, Campus Rent focuses on providing a convenient solution for students to share and rent items within their campus. It not only saves money but also encourages better utilization of resources and promotes a collaborative culture among students.

1.2.1 PROBLEM STATEMENT

In a college environment, students frequently require access to a wide variety of items such as academic books, electronic gadgets, laboratory tools, project equipment and other daily-use products. However these requirements are often temporary in nature. Purchasing such items for short-term use leads to unnecessary financial burden, especially for students who operate under limited budgets.

At the same time, it is commonly observed that many students already possess these items but they remain unused for extended periods after their initial purpose is fulfilled. This results in

inefficient utilization of available resources within the campus. Despite this imbalance between demand and availability there is no structured and reliable system that enables students to connect and share these resources effectively.

Currently, students rely on informal methods such as personal contacts, social media groups or word-of-mouth communication to borrow or lend items. These methods are often unorganized, time-consuming and lack transparency. They also raise concerns regarding trust, security, item condition and accountability, which discourages students from participating in such exchanges.

Furthermore the absence of a centralized platform makes it difficult to track transactions, manage bookings or ensure fair usage among students. This leads to missed opportunities for cost savings, collaboration and sustainable consumption within the campus community.

Therefore, there is a strong need to develop a secure, centralized and user-friendly web-based platform that facilitates item renting and sharing among students. The system should ensure proper authentication, transparent communication, easy listing of items and efficient management of rental transactions. The proposed solution, Campus Rent, aims to address these challenges by creating a digital ecosystem that promotes accessibility, trust and optimal utilization of resources within the college environment.

1.2.2 MOTIVATION

The motivation behind the development of the Campus Rent platform stems from the real-life challenges and inefficiencies experienced by students in their day-to-day academic and personal activities. During college life, students often encounter situations where they require certain items for a limited duration such as reference books before exams. tools for project work or electronic devices for temporary use. In such cases, purchasing these items is not always economical or practical.

At the same time it is evident that a large number of students possess items that are no longer in regular use. These underutilized resources represent an opportunity to create a system where items can be shared or rented, benefiting both the owner and the user. Recognizing this gap between demand and availability served as the primary inspiration for this project.

Another important motivating factor is the need to promote cost-effectiveness among students. By enabling renting instead of purchasing students can significantly reduce their expenses while still fulfilling their requirements. This approach not only benefits individuals but also contributes to a more sustainable and resource-efficient environment.

Additionally the project is driven by the idea of fostering a sense of collaboration and community within the campus. A platform like *Campus Rent* encourages interaction, trust-building, and mutual support among students. It transforms the campus into a more connected ecosystem where resources are shared rather than wasted.

From a technological perspective, the motivation also includes applying modern web development technologies such as React, Node.js, Express and MongoDB to solve a practical

real-world problem. This project provides an opportunity to design and implement a scalable, user-friendly and secure system that integrates frontend, backend and database functionalities effectively.

In conclusion, the Campus Rent project is motivated by the desire to solve practical student problems, reduce financial burden, improve resource utilization and build a collaborative campus environment through the use of innovative technology.

1.2.3 OBJECTIVES

The main objectives of this project are as follows:

1. To develop a centralized platform for renting and sharing items within the campus.
2. To help the users to easily list, browse and search for available items.
3. To implement secure user registration and authentication.
4. To provide smooth and transparent rental transactions.
5. To promote cost-effectiveness by encouraging renting instead of purchasing.

1.2.4 OVERVIEW OF METHODOLOGY

The methodology adopted for the development of the Campus Rent system follows a systematic and structured approach to ensure the successful design, implementation and deployment of the application. The development process is based on standard software engineering practices, focusing on clarity, efficiency and reliability at each stage.

Initially the project begins with the requirement analysis phase, where the needs and challenges faced by students are carefully studied. This includes understanding the problems related to accessing items, lack of a centralized platform, and issues of trust and transparency. Based on this analysis the functional and non-functional requirements of the system are defined clearly.

Following this, the system design phase is carried out where the overall architecture of the application is planned. The project is designed using a full-stack approach, where the frontend, backend and database components are structured to work together efficiently. The frontend is developed using React to provide an interactive and user-friendly interface. The backend is implemented using Node.js and Express to handle server-side logic, API creation and request management. MongoDB is used as the database to store user data, item details and transaction records in a flexible and scalable manner.

After designing the system, the implementation phase begins, where the actual coding of the application takes place. In this stage, features such as user registration and login, item listing, browsing, booking and rental management are developed and integrated. Proper validation and authentication mechanisms are implemented to ensure data security and system reliability.

Once the development is completed the system undergoes testing to identify and fix errors or bugs. Various testing methods are applied to ensure that the application functions correctly under different scenarios and provides a smooth user experience. This step is crucial to ensure that the system is stable, secure and efficient.

Finally, the deployment phase is carried out where the application is made accessible to users. The system is hosted on appropriate platforms allowing students to use it in real-time. Post-deployment, the system can be monitored and updated based on user feedback and future requirements.

Overall, the methodology ensures a well-organized development process, resulting in a robust, scalable and user-friendly Campus Rent platform that effectively addresses the needs of students within the campus environment.

1.2.5 SCOPE OF THE PROJECT

The scope of the Campus Rent project is focused on developing a web-based platform that facilitates the renting and sharing of items among students within a college campus. The system is designed to operate within a controlled and specific environment, primarily targeting students of a particular institution thereby ensuring a secure and trusted user base.

The project covers the development of core functionalities required for an efficient rental system. This includes user registration and authentication, allowing students to create accounts and securely access the platform. Users can list items they wish to rent out by providing necessary details such as item name, description, pricing and availability. Similarly, other users can browse through the available items, search based on their requirements and request rentals in a simple and convenient manner.

The system also includes features for managing rental transactions, such as booking items, tracking rental duration and handling item returns. It maintains proper records of all activities, ensuring transparency and accountability within the platform. Additionally the application provides a user-friendly interface that can be accessed across different devices, making it convenient for students to use anytime and anywhere.

However the scope of the project is limited to a campus-level implementation and does not initially include advanced features such as online payment integration, delivery logistics, or integration with external platforms. The focus remains on creating a simple, efficient and secure system for peer-to-peer renting within a closed community.

The project also emphasizes scalability, allowing future enhancements such as payment gateways, rating and review systems, real-time notifications, and expansion to multiple campuses. Thus, while the current scope addresses fundamental needs it also provides a foundation for further development and improvement.

In summary, the Campus Rent project aims to deliver a practical and effective solution for resource sharing within a campus, with a well-defined scope that balances functionality, simplicity and future growth potential.

1.2.6 ORGANIZATION OF THE REPORT

This project report is organized in a systematic manner to present the development and implementation of the Campus Rent system in a clear and structured form. Each chapter of the report focuses on a specific aspect of the project ensuring a logical flow of information and better understanding for the reader.

The first chapter provides an introduction to the project, including the background, problem statement, motivation, objectives, methodology and scope of the system. This chapter sets the foundation by explaining the purpose and significance of the Campus Rent platform.

The second chapter presents the literature review, where existing systems, research works and related technologies are discussed. It highlights the limitations of current approaches and establishes the need for the proposed system.

The third chapter describes the system analysis and design. It includes detailed explanations of system architecture, module functionalities and system flow explanation. This chapter provides a blueprint of how the system is planned and organized.

The fourth chapter focuses on the implementation of the system. It explains the technologies used, such as React for frontend development, Node.js and Express for backend processing and MongoDB for database management. It also describes the development of various modules and functionalities of the application.

The fifth chapter covers the testing and results of the system. It includes different testing methods used to ensure the reliability and performance of the application, along with the outcomes and observations.

The sixth chapter provides the conclusion and future scope of the project. It summarizes the overall work done, highlights the achievements of the system, and suggests possible enhancements and improvements for future development.

Overall, the report is structured in a way that guides the reader from the basic understanding of the problem to the final implementation and evaluation of the Campus Rent system ensuring clarity, coherence and completeness.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of Literature Review

The rapid advancement of digital technologies such as web applications, mobile computing, and cloud-based systems has transformed traditional ownership-based models into access-based systems. Rental platforms are a key component of this transformation, enabling users to access goods and services on demand without the need for permanent ownership. These systems contribute to cost efficiency, sustainability and optimized resource utilization [1], [2].

Earlier rental systems were limited to specific domains; however, modern research focuses on multi-category platforms and community-based ecosystems. The integration of technologies such as cloud computing, real-time databases, and secure payment gateways has further enhanced the scalability and efficiency of rental systems [3], [4]. This section examines the evolution of such platforms and identifies existing limitations that justify the need for a campus-focused rental solution.

2.2 Specific Domain Rental Platforms

Initial rental systems were designed to address specific sectors such as housing and transportation. Housing rental platforms allowed users to search and book properties online while ensuring trust through identity verification and secure transactions [5], [6]. These systems significantly improved accessibility and reduced manual efforts involved in traditional rental processes.

Similarly, vehicle rental systems were developed to streamline booking, availability tracking, and payment processes. These platforms incorporated authentication mechanisms and secure payment gateways to ensure reliability and reduce fraudulent activities [7], [8]. Mobile-based applications further enhanced user experience by providing real-time booking and location tracking features.

Despite their effectiveness, these systems suffered from major limitations. Their design was restricted to a single category making them unsuitable for users with diverse needs. Additionally, these platforms lacked flexibility and scalability preventing them from being adapted to broader use cases [9]. This limitation highlighted the need for more versatile systems capable of handling multiple categories within a unified framework.

2.3 Multi-Category Rental Platforms

To address the limitations of domain-specific systems, multi-category rental platforms were introduced. These platforms allow users to rent a variety of items such as electronics, furniture, tools, and household products through a single interface. Research indicates that such platforms improve convenience, accessibility and overall resource utilization [10]–[13].

These systems also incorporate advanced features such as rating systems, user feedback, and review mechanisms, which enhance trust and transparency among users. Studies show that these features play a crucial role in building user confidence and increasing engagement in collaborative consumption models [11], [12].

Furthermore, the concept of the sharing economy has significantly influenced the development of multi-category rental platforms. Research highlights that sharing-based models reduce resource wastage and promote sustainable consumption [13], [14]. These platforms also leverage modern web technologies to handle multiple users efficiently.

However, despite these advantages, multi-category systems still face challenges. They are generally designed for a broad audience and lack customization for specific communities such as students. Additionally managing a wide range of categories increases system complexity, which may impact performance and usability if not handled properly.

2.4 Campus Rental Platforms

Recent research has focused on developing rental platforms specifically tailored for academic environments. Students often require temporary access to items such as books, lab equipment, electronics and furniture. Due to financial constraints, purchasing these items is often impractical.

Campus-based rental platforms provide a solution by enabling students to share and rent resources within a closed and trusted environment [15]–[18]. These platforms ensure that all users belong to the same institution, thereby increasing trust and reducing risks associated with transactions.

Such systems typically include features such as institutional authentication, categorized listings, booking management, and secure payment systems. Studies suggest that these platforms significantly reduce costs for students while promoting collaboration and resource sharing within academic communities [16], [17].

However, existing research in this area is still limited. Many proposed systems lack scalability, real-time processing capabilities, and advanced security features. Additionally most implementations are either theoretical or lack practical deployment, indicating the need for a more comprehensive and efficient solution.

2.5 Supporting Technologies and System Efficiency

Modern rental platforms rely heavily on full-stack technologies to ensure performance, scalability, and security. Technologies such as Node.js are widely used due to their non-blocking architecture, which allows efficient handling of multiple concurrent requests [19]–[21].

Backend frameworks like Express.js and databases such as MongoDB provide flexible data handling and efficient backend processing, enabling real-time updates and secure transactions [22], [23]. These technologies are essential for building scalable and high-performance rental systems.

In addition, emerging technologies such as blockchain have been proposed to enhance transparency and trust in rental transactions. Blockchain ensures secure and tamper-proof record-keeping, reducing the chances of fraud and improving system reliability [24].

Despite these advantages the implementing such technologies introduces challenges such as increased system complexity, higher development costs, and the need for continuous optimization. Therefore, careful design and implementation are required to balance performance and usability.

2.6 Research Gap

The review of existing literature indicates that significant work has been done in the development of rental platforms, ranging from domain-specific systems such as housing and vehicle rentals to more advanced multi-category rental platforms. These systems have successfully demonstrated the use of digital technologies to facilitate resource sharing and improve accessibility.

However, most of the existing solutions are limited in scope. Domain-specific rental platforms focus only on a single category of services and lack flexibility, making them unsuitable for users who require access to diverse types of items. On the other hand, multi-category rental platforms, although more versatile, are generally designed for a broad audience and do not cater to the specific needs of particular communities such as students.

Furthermore, existing platforms often lack a secure and controlled environment tailored for campus use. Issues such as lack of user verification within a closed community, absence of trust mechanisms specific to academic institutions, and limited focus on affordability remain unaddressed. Many systems also do not provide a comprehensive solution that integrates features such as real-time booking, secure payment processing and efficient item management within a single platform.

Additionally the research on campus-specific rental systems is still limited and most proposed solutions are either conceptual or lack full-scale implementation using modern technologies. There is also a lack of focus on creating a user-friendly, scalable and secure system that can handle multiple categories of items while maintaining simplicity and efficiency.

Therefore there is a clear need for a dedicated platform that combines the advantages of multi-category rental systems with the trust, security and accessibility of a campus-based environment. The proposed *Campus Rent* system aims to address these gaps by providing a centralized, secure and efficient solution specifically designed for students, enabling better resource utilization and promoting a collaborative campus ecosystem.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Overview of Proposed Methodology

The proposed methodology for the *Campus Rent* system is designed to provide a structured, efficient and scalable solution for renting and sharing items within a campus environment. The system follows a full-stack development approach that integrates frontend, backend and database technologies to ensure seamless communication and real-time interaction.

The methodology focuses on solving key challenges such as lack of a centralized platform, inefficient resource utilization, and absence of trust in informal rental systems. By adopting a modular and layered architecture, the system ensures flexibility, security and ease of use for all stakeholders.

3.2 System Architecture Design

The architecture of the *Campus Rent* system is based on a three-tier model, which separates the application into distinct layers to improve maintainability and performance.

3.2.1 Presentation Layer (Frontend)

The presentation layer is responsible for user interaction and interface design. It is developed using React, which enables the creation of a dynamic and responsive user interface. Users can perform operations such as registration, login, item browsing, searching, booking and payment through this layer. The frontend communicates with the backend using RESTful APIs.

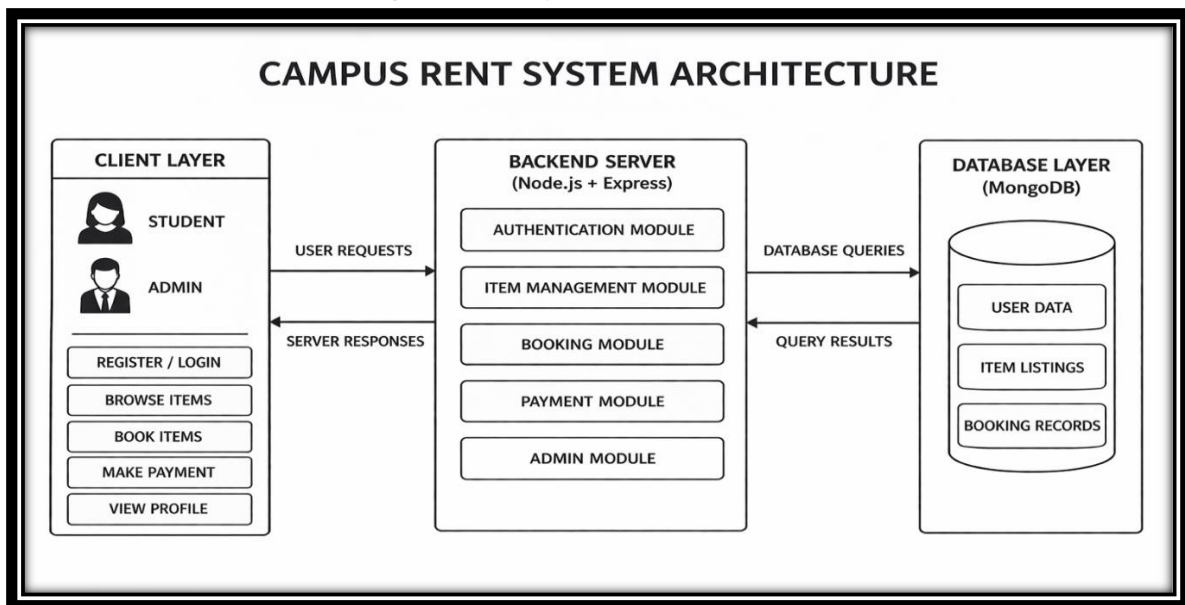
3.2.2 Application Layer (Backend)

The backend is developed using Node.js and Express.js, which handle all server-side operations. This layer processes user requests, applies business logic and ensures secure communication between the frontend and database. It also manages authentication, booking validation and transaction processing.

3.2.3 Data Layer (Database)

MongoDB is used as the database to store structured and unstructured data such as user profiles, item listings, booking records and transaction details. Its NoSQL nature allows flexibility in handling multiple categories of items and ensures efficient data retrieval.

Figure 3.1 System Architecture



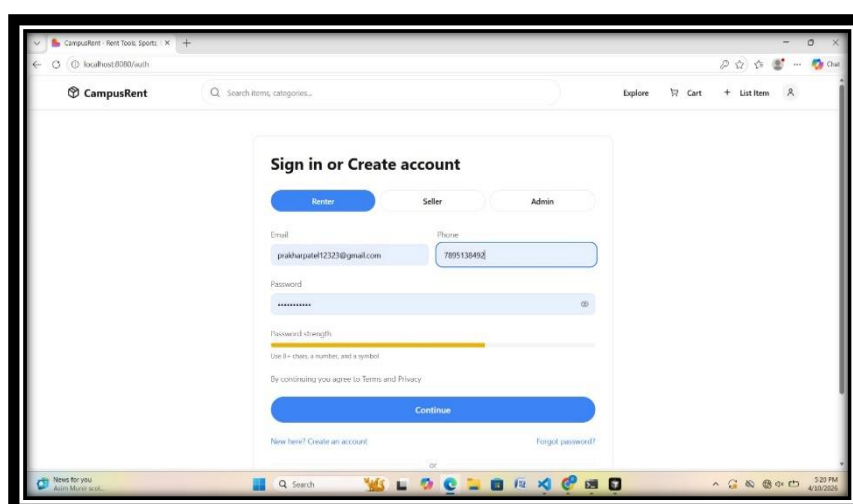
3.3 Functional Modules

The system is divided into several functional modules to ensure organized development and efficient operation.

3.3.1 User Authentication Module

This module handles user registration and login processes. It ensures that only verified users can access the platform. Passwords are encrypted and secure authentication mechanisms are implemented to protect user data.

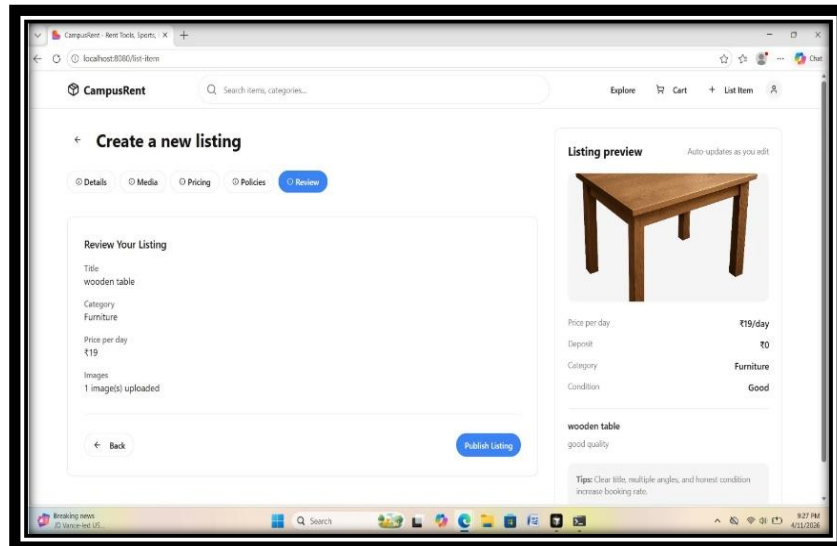
Figure 3.2 User Authentication Module



3.3.2 Item Listing and Management Module

Users can add, update or delete item listings. Each listing includes details such as item name, description, category, rental price and images. This module ensures proper management of items within the system.

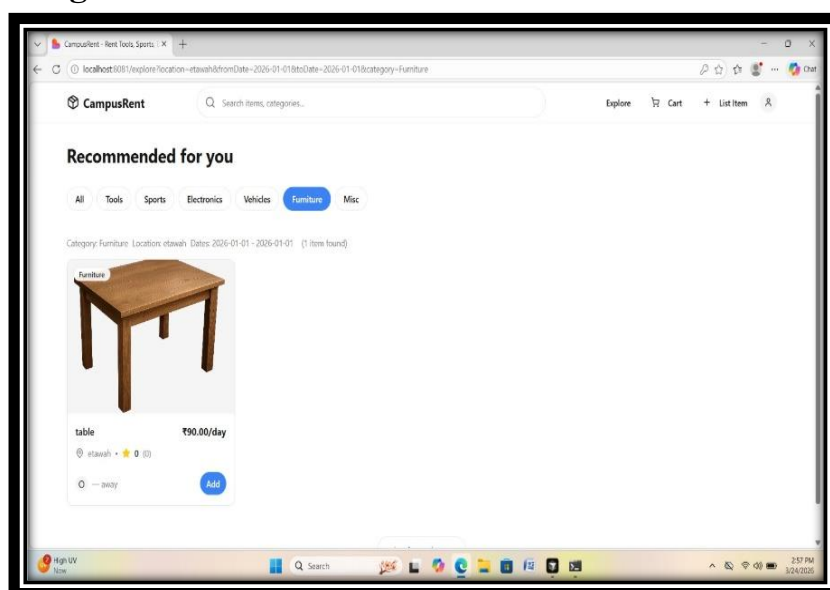
Figure 3.3 Item Listing Module



3.3.3 Search and Recommendation Module

This module allows users to search for items using keywords or browse through predefined categories. Advanced filtering options improve user experience and help users quickly find relevant items.

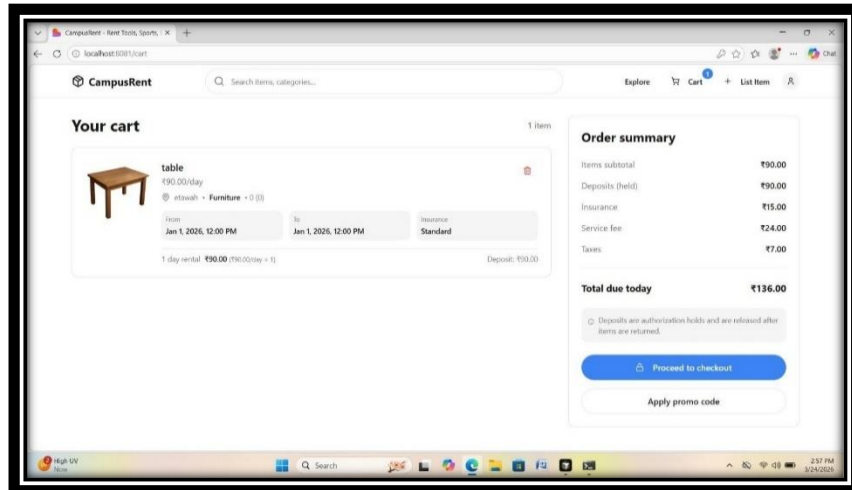
Figure 3.4 Search and Recommendation Module



3.3.4 Booking and Rental Module

Users can select items, specify rental duration and initiate booking requests. The system validates availability and updates booking status accordingly. This module ensures smooth and transparent rental transactions.

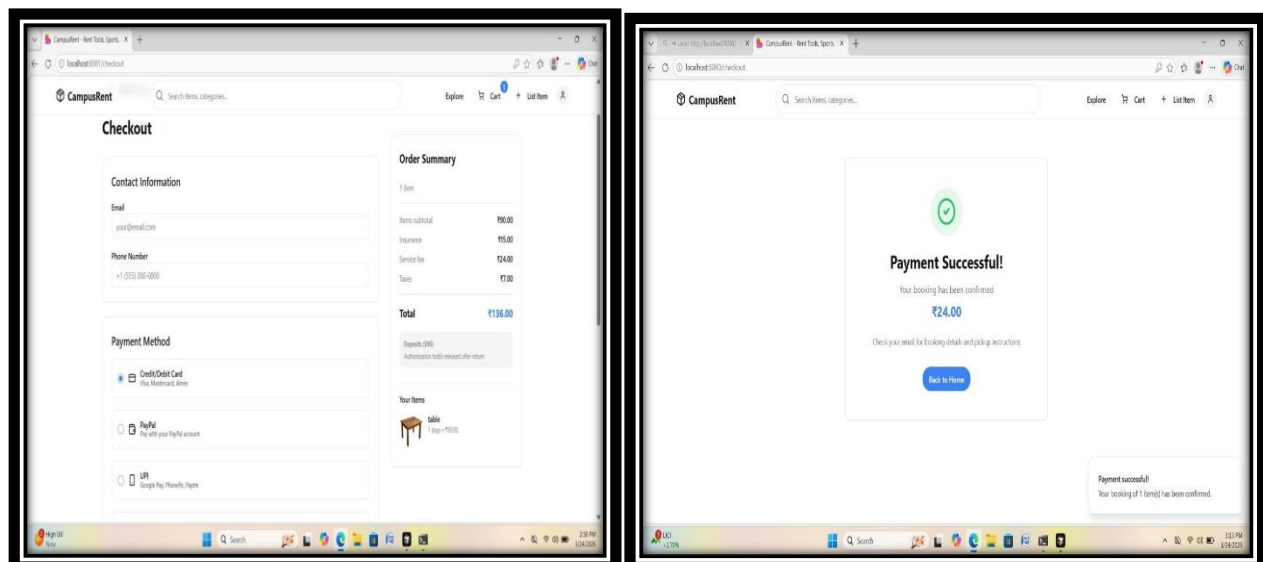
Figure 3.5 Booking Module



3.3.5 Payment Processing Module

A secure payment gateway is integrated to handle financial transactions. The system supports multiple payment methods and ensures safe and reliable payment processing.

Figure 3.6 Payment Processing Module



3.3.6 Admin Control Module

The admin module provides control over the entire system. It allows administrators to monitor users, manage listings, track transactions and handle suspicious activities.

3.4 System Flow Explanation

The system flow represents the step-by-step working of the *Campus Rent* platform:

3.4.1 Registration and Authentication Flow

The process begins when a user registers on the platform by providing necessary details. The system verifies the information and stores it securely in the database. After successful registration, the user logs in to access system features.

3.4.2 Item Listing Flow

Item owners upload details of items they want to rent. The backend processes the data and stores it in the database. These items are then displayed on the platform for other users.

3.4.3 Search and Selection Flow

Users search for required items using keywords or categories. The system retrieves relevant results from the database and displays them. Users can view item details before making a decision.

3.4.4 Booking Flow

After selecting an item, the user specifies the rental duration and sends a booking request. The system checks item availability and confirms the booking.

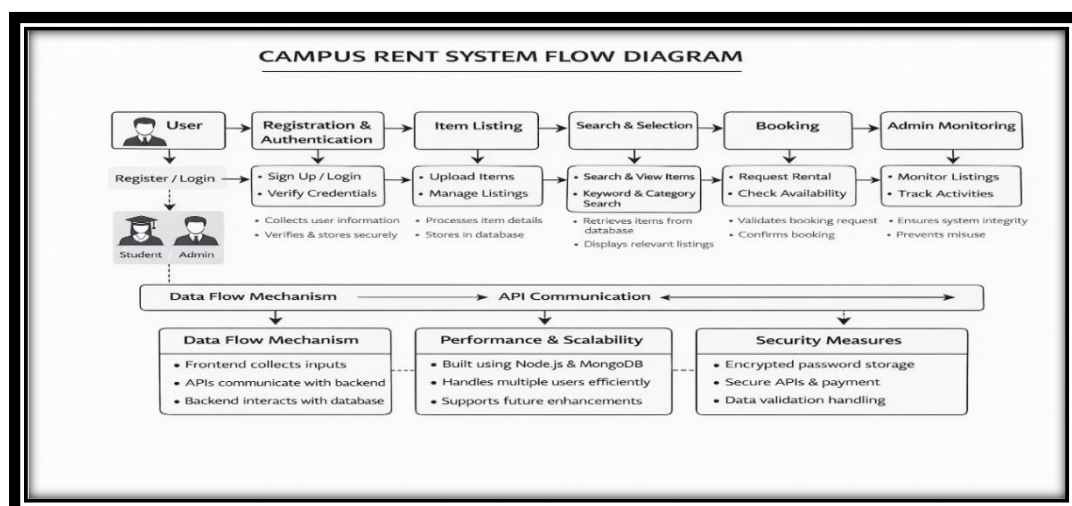
3.4.5 Payment Flow

Once the booking is confirmed, the user proceeds to payment. The payment gateway processes the transaction securely. After successful payment the booking is finalized.

3.4.6 Admin Monitoring Flow

The admin monitors system activities, including user behavior, item listings and transactions. This ensures system integrity and prevents misuse.

Figure 3.7 System Flow Diagram



3.5 Data Flow Mechanism

The system follows a structured data flow:

- User inputs are captured through the frontend.
- Requests are sent to the backend via APIs.
- The backend processes the request and interacts with the database.
- The database returns relevant data to the server.
- The server sends the response back to the frontend for display.

This continuous interaction ensures real-time updates and efficient system performance.

3.6 Performance and Scalability Considerations

The system is designed to handle multiple users simultaneously using Node.js which supports asynchronous processing. MongoDB ensures fast data retrieval and efficient storage. The modular architecture allows easy scalability, enabling future enhancements such as mobile applications, AI-based recommendations and advanced security features.

3.7 Security Measures

Security is a critical aspect of the system. The methodology includes:

- Encrypted password storage
- Secure authentication and authorization
- Protected API end points
- Secure payment integration
- Data validation and error handling

These measures ensure safe and reliable system operation.

3.8 Advantages of the Proposed Methodology

The proposed methodology offers several advantages:

- Provides a centralized and organized rental platform
- Supports multiple categories of items
- Ensures secure and transparent transactions
- Enhances user experience through a responsive interface
- Promotes efficient resource utilization within the campus

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Results

The *Campus Rent* system was successfully implemented and tested to evaluate its overall performance and effectiveness in a real-world campus environment. The system was examined across all major functionalities, including user authentication, item listing, search operations, booking process and payment integration.

During testing, it was observed that users were able to register and log in without any difficulty. The authentication system worked reliably, ensuring secure access to the platform. The item listing feature allowed users to upload item details easily, and all listed items were displayed correctly in their respective categories. This made it convenient for other users to browse and select items.

The search functionality performed efficiently by providing accurate results based on keywords and categories. Users were able to find items quickly which significantly reduced the time required to locate needed resources. The booking process was smooth and well-structured, allowing users to select items, choose rental duration and confirm their bookings without any confusion.

One of the key observations during testing was the speed of the system. The complete process—from user login to booking confirmation and payment—was completed in a short duration, typically within two minutes. The integration of the payment gateway ensured that transactions were processed securely and bookings were confirmed only after successful payment.

In addition to technical testing, a sample group of 20 students was involved in testing the system to evaluate its usability and practicality. The feedback collected from these users was largely positive. Most users found the platform easy to use, well-organized and helpful in quickly finding and renting items. They also appreciated the secure payment system and verified user environment, which increased their trust in the platform.

In terms of performance the system demonstrated the ability to handle multiple users simultaneously without noticeable delays. The backend efficiently managed concurrent requests, maintaining consistent response time and system stability.

Table 4.1: Showing Result and Evaluation

Evaluation Aspect	Test Scenario / Criteria	Observed Result	Interpretation
Authentication	User registration and login	Users successfully logged in	Secure and reliable access
Item Management	Add, update, and display items	Items correctly stored and displayed	Accurate data handling
Search & Navigation	Search by keyword and category	Relevant and fast results	User-friendly system
Booking Process	Complete booking with duration selection	Booking completed without errors	Smooth workflow
Payment System	Online transaction via gateway	Payment processed successfully	Secure and dependable
Performance Speed	Time to complete full booking	Completed within ~2 minutes	Fast processing
System Response	Response time during operations	Few hundred milliseconds	Efficient performance
Scalability	Multiple users accessing system	200+ users handled simultaneously	Scalable architecture
User Testing	Testing with 20 students	Positive feedback and easy usability	High user acceptance
Security	Data protection and authentication	Encrypted passwords and secure APIs	Strong security implementation

4.2 Discussion

The results obtained from testing highlight the effectiveness of the *Campus Rent* system in addressing the challenges faced by students in renting items within a campus. The platform provides a structured and centralized solution, eliminating the need for informal and unorganized methods such as social media groups or manual communication.

One of the major strengths of the system is its simplicity and ease of use. The feedback from the 20-user sample group clearly indicates that even first-time users were able to understand and operate the system without difficulty. The categorized structure and simple navigation played a key role in improving the overall user experience.

Another important aspect is the speed and efficiency of the system. The ability to complete a full rental process in a short time makes it highly practical for everyday use. The backend architecture ensures that multiple users can access the system simultaneously without affecting performance, which is essential for real-world deployment.

The integration of secure payment and authentication mechanisms significantly improves user trust. The sample users reported feeling more confident using the platform compared to informal renting methods, mainly because of the structured system and secure transactions.

The system also promotes better resource utilization by allowing students to share unused items. This not only reduces unnecessary expenses but also encourages a collaborative environment within the campus.

However there is still scope for improvement. Features such as real-time notifications, user ratings, and a dedicated mobile application can further enhance the system. Despite these minor limitations, the system proves to be a reliable and practical solution.

Overall the *Campus Rent* system successfully fulfils its purpose by providing a secure, efficient and user-friendly platform. The positive feedback from the sample users further validates its effectiveness and potential for wider adoption.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The *Campus Rent* system was developed to address a practical and commonly faced problem among students—accessing items that are required only for a short duration. In many cases students either purchase such items at a high cost or depend on informal methods like friends or social media groups, which are often unreliable and unorganized. The proposed system provides a structured and efficient solution to this problem by introducing a centralized platform where students can rent and share items within their campus.

The system was successfully designed and implemented using modern full-stack technologies. It integrates a responsive frontend, a secure backend, and a flexible database to ensure smooth interaction between users and the system. Key features such as user registration and authentication, item listing and categorization, search functionality, booking management and secure payment integration were developed and tested thoroughly.

During testing, the system demonstrated strong performance in terms of speed, reliability, and usability. Users were able to complete the entire process—from login to booking and payment—in a short time, making the platform highly efficient for daily use. The system was also able to handle multiple users simultaneously without any noticeable delay, which indicates good scalability.

Another important outcome of this project is the positive user response. The feedback from the sample group showed that users found the system easy to use, well-organized, and helpful in saving both time and money. The structured design and secure transaction system increased user confidence which is very important in peer-to-peer platforms.

In addition to technical success the system also contributes to a larger purpose. It promotes the idea of sharing resources instead of owning them unnecessarily, which helps reduce waste and encourages a more sustainable lifestyle within the campus. By enabling students to make better use of available resources, the platform creates a more collaborative and supportive environment.

Overall, the *Campus Rent* system successfully achieves its objectives. It provides a practical, reliable and user-friendly solution for campus-based renting and has the potential to make a meaningful impact on student life.

5.2 Future Scope

Although the current system performs well and fulfills its primary objectives, there are several opportunities for further improvement and expansion that can enhance its functionality and user experience.

One of the most important future enhancements is the development of a dedicated mobile application. While the current system works well as a web application, a mobile app would provide greater convenience and accessibility. Users would be able to receive instant notifications for booking updates, payment confirmations and new item listings, making the platform more interactive and user-friendly.

Another useful feature that can be added is a rating and review system. This would allow users to share their experiences after each transaction, helping others make informed decisions. It would also build trust within the platform by encouraging responsible behavior among users.

The system can also be improved by introducing a recommendation system. By analyzing user behavior such as search history and previous bookings the platform can suggest relevant items to users. This would make the system smarter and enhance the overall user experience.

From a security perspective, advanced technologies such as blockchain can be explored in the future. Blockchain-based smart contracts can ensure that all rental agreements are transparent and cannot be altered which would further increase trust and security, especially for high-value transactions.

Another area of improvement is the integration of real-time communication features, such as chat or messaging between users. This would allow renters and owners to communicate directly within the platform making the process more efficient and reducing dependency on external communication tools.

The system can also be expanded beyond a single campus to multiple institutions. By connecting different colleges or universities, the platform can create a larger network of users, increasing the availability of items and improving overall usability. This expansion can turn the platform into a large-scale rental ecosystem.

Additionally the features like digital ID verification, automated reminders for item return and analytics dashboards for tracking usage can further enhance the system's functionality and reliability.

In conclusion, the *Campus Rent* system has strong potential for future growth and development. With the addition of advanced features and expansion to a larger user base it can evolve into a comprehensive and widely adopted platform for resource sharing and renting.

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APPENDIX

RESEARCH PAPER

Proceedings of the 5th International Conference on Sentiment Analysis and Deep Learning (ICSADL-2026)
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Campus Rent: A Full-Stack Web Application for Item Renting in University Environments

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Abstract— Campus Rent is a full-stack web platform that powers item rentals on university campuses in a peer-to-peer manner. It aims to replace fragmented owner-centricism with an access-based model, enabling students to rent electronics, tools, sports equipment, furniture, and more seamlessly on the portal. It combines a responsive frontend of React with a Node.js/Express-based backend and MongoDB as the database. Payment integration has been performed with the Razor pay gateway, which facilitates card-based, UPI-based, or cash-on-delivery-based transactions. The key functionalities of the site are user authentication; posting of items with categorization, real-time booking, and an admin dashboard for community moderation. It showed that rentals could be completed in under two minutes and handled hundreds of users at the same time with very little delay during testing. At the campus level, Campus Rent will advocate "rent-instead-of-owning" to ensure affordability and sustainability. In the future, further development could add functions such as blockchain contracts for convenience and trust-building.

Keywords— Campus Rent, Item Renting, peer-to-peer rental, sharing economy, full-stack web development, campus sustainability

I. INTRODUCTION

Today, people don't always want to own everything that they use. For many products, being able to borrow them for a while is akin to having a store that lends things out and takes them back. Peer-to-peer rental websites and apps are facilitating this shift by enabling those with extra stuff to connect with others who want it. That way, items such as tools, gadgets or sports gear no longer just gather dust but can be conveniently shared. This trend is based on the move toward access-based consumerism, and the fact that use when wanted is more important than ownership [1-2]. University students are a great example of those who can take advantage of access-based approaches. While they are away at school, there may be things that they need for the semester, such as electronics, lab equipment, sports gear or some basic furniture for their dorm [3].

Campus Rent is a student-focused solution to an all-too-common campus problem: not being able to get need, and instead buying it for exorbitant prices for such a short amount of time [4]. In each campus, there's a constant need for calculators, lab equipment, books, power banks, cycles, project kits, or other supplies for just a few days to spend in

every college or university. It's not as if these products are inexpensive to buy outright, either, and that amount is likely enough of a stretch for students trying to watch their pennies. In any case, many other students already possess these things; however, they have barely given orders to use them [5-6]. Campus Rent is bridging this gap by creating a safe online marketplace within the university through which students can lend and borrow to one another. It's an elegantly simple idea that leverages and shares resources with the campus, minimizes theft and nets savings. Since use of the site is limited only to those verifiable as real users and with a true student account, everyone on it shares being from the same institutional system, and can therefore warrant trust in one another at some higher level. In this way, not only does it operate as a marketplace of transactions, but it also promotes a culture of co-operation and trustworthiness between students. Look at Campus Rent, for example, which does nothing more than turn college kids' garbage into a useful resource and one that can save students dollars [7].

The backend is 'multi-threaded', which means that author can serve more than one user at a time. The user profile, listed item, transaction record, payment information and the rating will be kept in a versatile document-oriented database system such as MongoDB [8-9]. It uses MongoDB for quick retrieval of data and would work for the same after adding the claims to the database. Strong authentication will be present so that user account protection is enforced. Overall, React & MongoDB make a great tech stack to keep the product fast, effective and would be relatively easy to maintain while developing new features [10-11]. The admin can monitor the new user accounts and check for suspicious use.

II. LITERATURE REVIEW

Recently Peer-to-peer digital enabled platforms, alongside web technology, mobile devices and cloud systems have facilitated recent growth in this area of modern rental operations. Previous studies widely covered rental scenario in limited domains such as transportation, housing and tourism. More recently, the focus of attention has switched to broader, more community focused rental models.

When online rental companies started, they generally offered only one type of service. Typically, a website would cater to either car bookings, property rentals or accommodation, but not all together. The author [11-12]

developed a housing rental portal in which users can book homes online, verify the profiles of one another, and communicate through their system. In their research, they concluded that the check-in process can feel safe and trustworthy only if it checks both owner and tenant information as well. Another author [13-14] proposed a car rental system with strengthened login and authentication procedures by which to decrease fake bookings as well as payment fraud. Development of a mobile application for a vehicle rental system. They wanted to make the app very easy to navigate and facilitate payments that are so quick that it feels simple and convenient. It eventually failed while it was limited to vehicles and did not address the broader shared items or other product domains [15]. The author [16] developed a full-stack car rental application contains a responsive page with integrated payment. They were able to show the efficiency of Node.js in handling high-volume parallel user interactions. This proves that full-stack approaches are fitting for high-demand rental platforms. However, all these systems lacked mixed-category item listings or verification tailored for a university-restricted user base. The researchers had also proposed a MERN-based system called "Flexible Sharing and Leasing" for vehicle rentals. They found that because Node.js uses a non-blocking structure, the platform can handle many users at the same time without slowing down [17-18].

The recent work focuses on multiple item-type rental platforms, rather than single-service ones. The author [19-20] proposed "Online Rental Things," a general-purpose website which allows users to create listings of products to be used, such as electronics, furniture, tools, and household items. Since it allows multiple categories, the system advances the share-and-borrow process of what one has and what one needs. Allowing the mentioned types of listings, the platform attempts to create trust among the users and motivate a sense of community and collaboration. The proposed an Android-based marketplace. This is designed to assist users in renting books, electronics, and other small furniture. Their work showed that user ratings and reviews make peer-to-peer exchanges trustable. The author [21-22], explored various models of rental platforms to identify the key challenges developers face. Intelligent Instrument Interface, a tool that allows music students and teachers to connect directly through peer-to-peer technology. This idea shows how similar systems can be used in universities for sharing or renting academic items. Together, these studies show that secure and user-friendly platforms are becoming more important. Although there is not much research about specific university rental platforms, a number of studies explore the solutions for academic communities. The author [23-25] suggested the Smart Student Rental Platform with the aim of maintaining the affordability and sustainability of living for students.

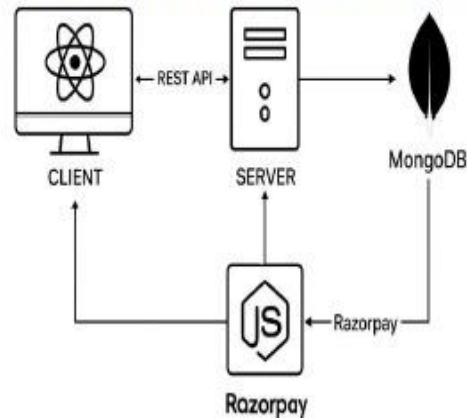
III. METHODOLOGY

Students are running into a lot of problems when renting things on or near campus. First of all, the rentals are not centralized in one place. Informal groups, as well as online social media and even community notice boards, are among them. It takes a lot of time and work to figure out what they need. Campus Rent removes these barriers and creates a safe online community for students. It pools all the types of rentals together [26-28].

System Design and Architecture:

Campus Rent is designed as a simple, three-piece web system: the part that the students see and use; the server that works in the background to get the job done and the database where all of the information is stored.

Client Layer (Frontend): The student UI is a web portal built in React that works great across phones and laptops. Although this interface, in which a user logs on, looks at items by category (books, lab equipment or the like), checks if something's available, requests it to borrow and pays for it



all in the one operation [29-30]. High level architecture (as shown in figure 1 above) is client browsers talks to Node/Express server and the server talks to MongoDB for data and Razor pay for payments.

Fig. 1. System Architecture of Campus Rent [30]

Server Layer (Back-end): The student UI is a React web frontend. This is where the rubber meets the road. It uses a Node.js server with the Express framework for serving the frontend and a database. The server has routes for signing up, logging in, finding an item, scheduling an appointment and checking out payment. Because Node.js, a single thread can support many thousands of input-output connections, and not having to start a new thread for each concurrent student is very beneficial since the server can efficiently utilise the Database Layer [31-32].

Database (Data Layer): It is utilized as a repository by MongoDB to persist records for users, postings, reservation requests and transactions. Its "schema-less" document model applies to the differing ways in which items are organized by category. Schema definitions and queries are performed through Mongoose ODM [33-34].

A. Module Functionality:

The system comprises the following key modules as shown in Fig. 2:

Authentication: User registration, login, password hashing, and session.

Item Catalog & Search: CRUD operations for item listings; browse by category or search by keywords.

Booking: Create rental requests, select data range and confirm.

Payment: Integrate Razor pay to process transactions securely.

Administration: Dashboards for monitoring users, items, and rental statistics.

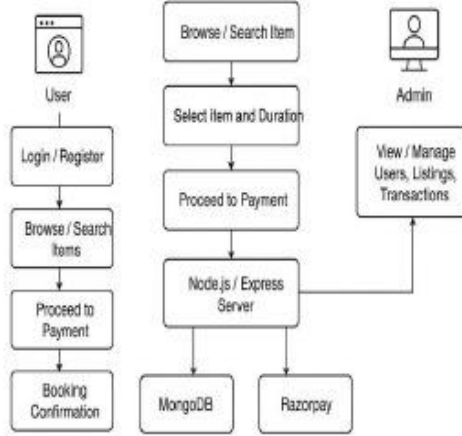


Fig. 2. Module Functionality

IV. PROPOSED SYSTEM ARCHITECTURE

This section discussed how each module was implemented using modern web technologies:

Authentication Module: Students register with their university email address and a password. Passwords are hashed-preferably with crypt and JSON Web Token is issued at login. MongoDB stores the user profiles with a unique student ID and email for verification [35].

Item Listing Module: Users can create a listing by giving information such as name, description, category, rental rate, condition, and images. Then, the server will store each listing in a MongoDB collection called Items. Users are allowed to make changes or delete their own items [36].

Category Browsing and Search: At the frontend, items are categorized into predefined categories, including Electronics, Tools, Sports, Furniture, Vehicles, Others. The various routes in Express query MongoDB for items that match with category or search terms; pagination for large result sets is also used [37].

Booking Module: Upon selecting an item, a renter specifies the dates/duration for which renting the item. The frontend sends this request to the server to create a provisional booking record. No capture of payment yet [38].

Payment Module: The server incorporates Razor pay's Node.js SDK. A Razor pay order is created for every booking by the server, which then responds with the ID of that order to the client. The frontend opens the Razor pay checkout. Once the payment is made successfully, Razor pay notifies the server through webhook. The server verifies the signature of the payment and marks the booking as confirmed in MongoDB [39].

Admin Dashboard: A special admin account can log in to a dashboard page. It displays real-time tables of all users, all listed items, and all completed transactions. Admins can also deactivate suspicious listings or users. Data is queried from MongoDB and displayed using simple HTML tables and

Figures 2. Through its implementation, the client and server communicated via a RESTful JSON API. Validation of incoming data occurs at the front-end form and backend route levels to prevent invalid data. For example, all required fields in a booking - dates, and item ID are checked on the server side to ensure consistent state.

TABLE 1. SYSTEM MODULES AND FUNCTIONALITIES

Module	Description
Authentication	Handles user registration, login, password encryption, and session management for secure access.
Item Catalog & Search	Allows CRUD operations for item listings, including browsing by category or keyword search.
Booking	Enables users to select items, choose rental duration, and create booking requests.
Payment	Integrates Razor pay gateway for secure online transactions through UPI, cards, or net banking.
Administration	Provides admin dashboard for managing users, listings, transactions, and system statistics.

V. RESULT & DISCUSSION

Functional correctness, time-constrained speed and a few Ux aspects of the system were checked. The test confirmations are: module-core: Account-registration-and-login, item-listing, booking-process-and-payment is good. In the test scenario, there was a whole booking test from login to confirmation of payment in approximately two minutes and author had a pretty good robust end-to-end pipeline. Performance-wise, Node.js (event-driven) back-end served concurrent requests very easily. The response time is in the scale of a few hundred milliseconds for systems that tens of thousands of users, as required by industrial rental platforms. In simulated testing, the Express.js server handled few hundreds of requests at the same time very well, keeping very similar latency even that a scalable web app for doing so. Future proposed experimentation author would like to conduct scaled-up stress testing so as to establish more thoroughly the system's robustness. A usage test with 20 students has revealed a strong acceptance in the user group. "Easy to navigate system, by category, with user friendly helping tool tips," wrote one rater about the user-friendly design of the booking feature. The integrated paywall and member verification served as a sort of perceived trust blanket. Believing in the platform was more acceptable for people seeking safety than ad hoc, peer-to-peer rental food swaps. In other words, the system architecture is tight and cheap in terms of costs and resources.

TABLE 2. RESULTS AND EVALUATION

Test Parameter	Observation	Remarks
Functional Testing	Core modules—including login, item posting, booking, and payments—worked smoothly during testing.	System functions verified successfully.

Performance	Average booking was completed in under 2 minutes, and the platform handled over 200 simultaneous users.	Scalability targets achieved.
User Feedback	A pilot group of 20 students reported positive experiences highlighting ease of use and reliability.	Higher user satisfaction than traditional methods.
Security	User passwords were protected using encrypt, and all payments were securely processed through the Razor pay SDK.	Follows recommended security standards.
Usability	The interface was rated as easy to navigate and worked well on various devices.	Strong potential for widespread adoption.

VI. CONCLUSION

Campus Rent is made for those who are on campus, and offers a solution to rent & share everything among the members of this community. Renting has never been so simple through a well-managed tenant service and secure digital instant bad credit payday loans transfer processes. The author does it with ID-verified user accounts, so students can browse intuitively. The revised model is efficient because it encourages re-use and thus decreases waste by the shared use of these resources in academic environments. Over and above what already exists, Campus Rent has a high potential for development. An automatic-recommendation system might examine modules already rented, for instance, and recommend new ones to pick up or rent. IoT-enabled smart lockers or digital locks can also be included to facilitate hassle-free pickup and return, where the user may access their rental in a socially distanced manner. Expensive rentals might involve blockchain-powered smart contracts to ensure that nobody changes the terms of the lease or billing information. Right now, the platform operates via a web dashboard. But a new mobile app would make this process more user-friendly and accessible, with push notifications that are updated in real-time, better interaction between buildings around campus, and a unique user experience for each. Formal usability testing would more definitively hone this interface to fit students. Campus Rent is, in its essence, a brand-new way to fill the gap of campus services that lack serious competitiveness and trustworthy rental service in multiple categories within a close academic community. Beyond testing for systems' feasibility and usability, such an implementation of the setup could also identify cost savings.

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Abstract:

Campus Rent is a full-stack web platform that powers item rentals on university campuses in a peer-to-peer manner. It aims to replace fragmented owner-centrism with an access-based model, enabling students to rent electronics, tools, sports equipment, furniture, and more seamlessly on the portal. It combines a responsive frontend of React with a Node.js/Express-based backend and MongoDB as the database. Payment integration has been performed with the Razor pay gateway, which facilitates card-based, UPI-based, or cash-on-delivery-based transactions. The key functionalities of the site are user authentication; posting of items with categorization, real-time booking, and an admin dashboard for community moderation. It showed that rentals could be completed in under two minutes and handled hundreds of users at the same time with very little delay during testing. At the campus level, Campus Rent will advocate "rent-instead-of-owning" to ensure affordability and sustainability. In the future, further development could add functions such as blockchain contracts for convenience and trust-building.

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